

3.11 A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If x is the number of defective sets purchased by the hotel, find the probability distribution of X . Express the results graphically as a probability histogram.

3.18 A continuous random variable X that can assume values between $x = 2$ and $x = 5$ has a density function given by $f(x) = 2(1 + x)/27$. Find

(a) $P(X < 4)$;

(b) $P(3 \leq X < 4)$.

3.20 For the density function of Exercise 3.18, find $F(x)$, and use it to evaluate $P(3 \leq X < 4)$.

3.13 The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct the cumulative distribution function of X .

3.39 From a sack of fruit containing 3 oranges, 2 apples, and 3 bananas, a random sample of 4 pieces of fruit is selected. If X is the number of oranges and Y is the number of apples in the sample, find

- (a) the joint probability distribution of X and Y ;
 - (b) $P[(X, Y) \in A]$, where A is the region that is given by $\{(x, y) \mid x + y \leq 2\}$.
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3.49 Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

$f(x, y)$		x		
		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

- (a) Evaluate the marginal distribution of X .
- (b) Evaluate the marginal distribution of Y .
- (c) Find $P(Y = 3 \mid X = 2)$.

Find the covariance of the random variables X and Y of Exercise 3.49 on page 106.

3.57 Let X , Y , and Z have the joint probability density function

$$f(x, y, z) = \begin{cases} kxy^2z, & 0 < x, y < 1, \ 0 < z < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find k .
 - (b) Find $P(X < \frac{1}{4}, Y > \frac{1}{2}, 1 < Z < 2)$.
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3.62 An insurance company offers its policyholders a number of different premium payment options. For a randomly selected policyholder, let X be the number of months between successive payments. The cumulative distribution function of X is

$$F(x) = \begin{cases} 0, & \text{if } x < 1, \\ 0.4, & \text{if } 1 \leq x < 3, \\ 0.6, & \text{if } 3 \leq x < 5, \\ 0.8, & \text{if } 5 \leq x < 7, \\ 1.0, & \text{if } x \geq 7. \end{cases}$$

- (a) What is the probability mass function of X ?
 - (b) Compute $P(4 < X \leq 7)$.
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4.1 The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given in Exercise 3.13 on page 92 as

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Find the average number of imperfections per 10 meters of this fabric.

4.7 By investing in a particular stock, a person can make a profit in one year of \$4000 with probability 0.3 or take a loss of \$1000 with probability 0.7. What is this person's expected gain?

4.12 If a dealer's profit, in units of \$5000, on a new automobile can be looked upon as a random variable X having the density function

$$f(x) = \begin{cases} 2(1 - x), & 0 < x < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find the average profit per automobile.

4.20 A continuous random variable X has the density function

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = e^{2X/3}$.

4.34 Let X be a random variable with the following probability distribution:

x	-2	3	5
$f(x)$	0.3	0.2	0.5

Find the standard deviation of X .

4.58 The total time, measured in units of 100 hours, that a teenager runs her hair dryer over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Use Theorem 4.6 to evaluate the mean of the random variable $Y = 60X^2 + 39X$, where Y is equal to the number of kilowatt hours expended annually.

5.12 A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within the state. What is the probability that fewer than 4 of the next 9 vehicles are from out of state?

5.26 Assuming that 6 in 10 automobile accidents are due mainly to a speed violation, find the probability that among 8 automobile accidents, 6 will be due mainly to a speed violation

- (a) by using the formula for the binomial distribution;
 - (b) by using Table A.1.
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5.57 On average, a textbook author makes two word-processing errors per page on the first draft of her textbook. What is the probability that on the next page she will make

- (a) 4 or more errors?
 - (b) no errors?
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6.8 Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

- (a) the normal curve area to the right of $x = 17$;
 - (b) the normal curve area to the left of $x = 22$;
 - (c) the normal curve area between $x = 32$ and $x = 41$;
 - (d) the value of x that has 80% of the normal curve area to the left;
 - (e) the two values of x that contain the middle 75% of the normal curve area.
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6.11 A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
 - (b) what is the probability that a cup contains between 191 and 209 milliliters?
 - (c) how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
 - (d) below what value do we get the smallest 25% of the drinks?
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6.19 A company pays its employees an average wage of \$15.90 an hour with a standard deviation of \$1.50. If the wages are approximately normally distributed and paid to the nearest cent,

6.21 The tensile strength of a certain metal component is normally distributed with a mean of 10,000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. Measurements are recorded to the nearest 50 kilograms per square centimeter.

- (a) What proportion of these components exceed 10,150 kilograms per square centimeter in tensile strength?
 - (b) If specifications require that all components have tensile strength between 9800 and 10,200 kilograms per square centimeter inclusive, what proportion of pieces would we expect to scrap?
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